Hyperbolas

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Standard form

For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

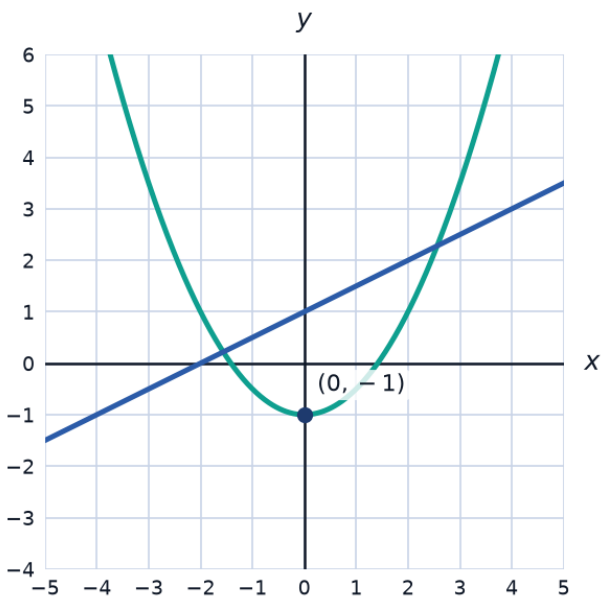
a the vertices

b the foci

c the equations of the asymptotes

2 [Foundational] • Target: Standard form

Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.



3 [Foundational] • Target: Standard form

Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

a Write the equation in standard form by completing the square.

b Identify the conic and state its center.

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4 [Foundational] • Target: Identifying conics

Identify each conic section from its equation:

a $x^2 + y^2 = 49$

b $\frac{x^2}{9} + \frac{y^2}{25} = 1$

c $y^2 - 4x^2 = 16$

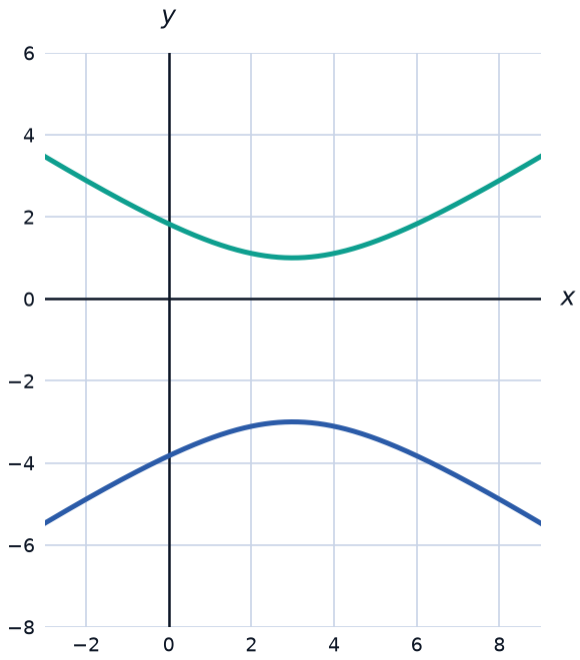
d $y = 3x^2 - 6x + 1$

5 [Foundational] • Target: Identifying conics

A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

6 [Foundational] • Target: Hyperbolas centered away from the origin

For the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:



a the center

b the vertices

c the direction of opening

7 [Foundational] • Target: Hyperbolas centered away from the origin

Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eccentricity and applications

Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

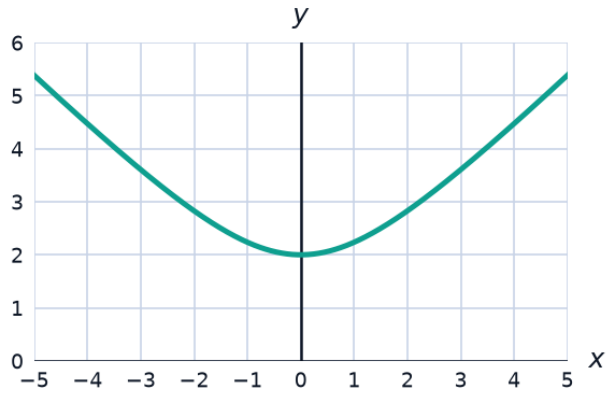
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9 [Intermediate] • Target: Eccentricity and applications

A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

10 [Intermediate] • Target: Graphing a hyperbola branch

The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate y at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.



11 [Intermediate] • Target: The hyperbola's defining property

State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.

12 [Intermediate] • Target: Comparing all four conic sections

State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \ldots = 0$.

13 [Intermediate] • Target: Finding the equation from asymptotes and a point

A hyperbola centered at the origin has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

14 [Intermediate] • Target: Applications: LORAN navigation

Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

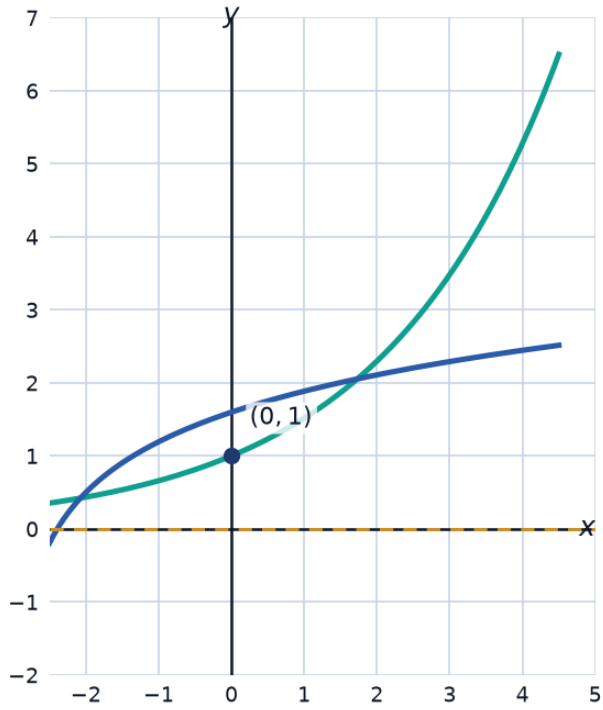
15 [Challenging] • Target: Identifying conics • Cross-Lesson: 9.03 The Binomial Theorem

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

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16 [Challenging] • Target: Hyperbolas centered away from the origin • Cross-Lesson: 1.01 Functions and Function Notation

Solve the logarithmic equation $\log_2(x+3) + \log_2(x-3) = 4$ for all real solutions x. State any extraneous roots.



17 [Challenging] • Target: Eccentricity and applications • Cross-Lesson: 1.02 Domain and Range

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

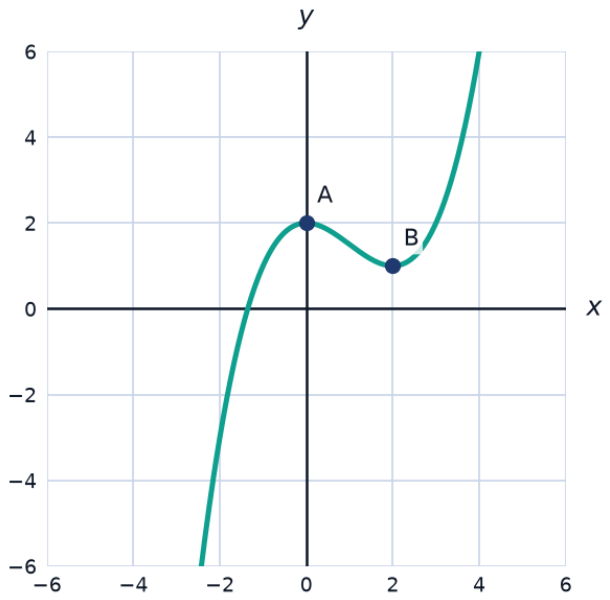
18 [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 1.03 Average Rate of Change

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n-1) = n^2$.

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19 [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



20 [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 2.02 Radian Measure and Arc Length

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Equations of Hyperbolas

Find the vertices and orientation of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. Horizontal, vertices at $(\pm 4, 0)$

b. Vertical, vertices at $(0, \pm 3)$

c. Horizontal, vertices at $(\pm 5, 0)$

d. Vertical, vertices at $(0, \pm 4)$

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22 [Foundational] • Target: Standard Equations of Hyperbolas

Find the vertices and orientation of $\frac{y^2}{25} - \frac{x^2}{144} = 1$.

a. Vertical, vertices at $(0, \pm 5)$

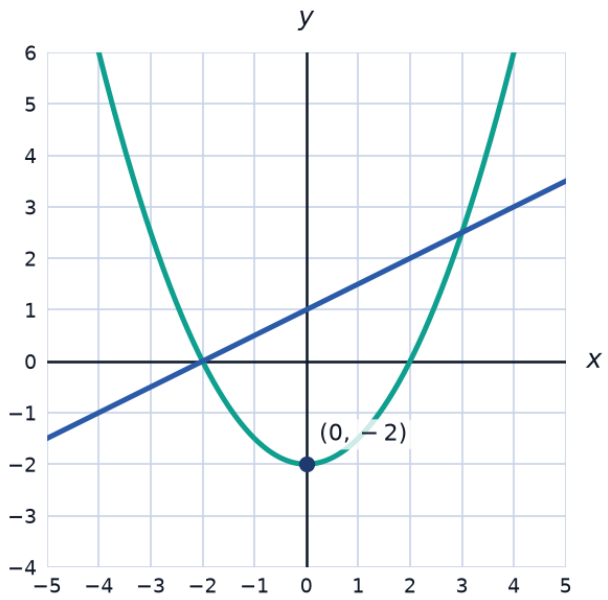
b. Horizontal, vertices at $(\pm 12, 0)$

c. Vertical, vertices at $(0, \pm 13)$

d. Horizontal, vertices at $(\pm 5, 0)$

23 [Foundational] • Target: Standard Equations of Hyperbolas

Find the length of the transverse axis and conjugate axis for $\frac{(x-1)^2}{36} - \frac{(y+2)^2}{64} = 1$.



a. Transverse $= 12$, Conjugate $= 16$

b. Transverse $= 6$, Conjugate $= 8$

c. Transverse $= 16$, Conjugate $= 12$

d. Transverse $= 72$, Conjugate $= 128$

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24 [Foundational] • Target: Asymptotes of Hyperbolas

State the equations of the asymptotes for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

a. $y = \pm\frac{b}{a}x$

b. $y = \pm\frac{a}{b}x$

c. $y = \pm\frac{b^2}{a^2}x$

d. $y = \pm x$

25 [Foundational] • Target: Asymptotes of Hyperbolas

Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{4}{3}x$

b. $y = \pm\frac{3}{4}x$

c. $y = \pm\frac{16}{9}x$

d. $y = \pm\frac{5}{3}x$

26 [Foundational] • Target: Asymptotes of Hyperbolas

Find the asymptotes of the vertical hyperbola $\frac{y^2}{4} - \frac{x^2}{9} = 1$.

a. $y = \pm\frac{2}{3}x$

b. $y = \pm\frac{3}{2}x$

c. $y = \pm\frac{4}{9}x$

d. $y = \pm 2x$

27 [Foundational] • Target: Foci and Eccentricity of Hyperbolas

What is the relationship between a, b, and focal distance c for a hyperbola?

a. $c^2 = a^2 + b^2$

b. $c^2 = a^2 - b^2$

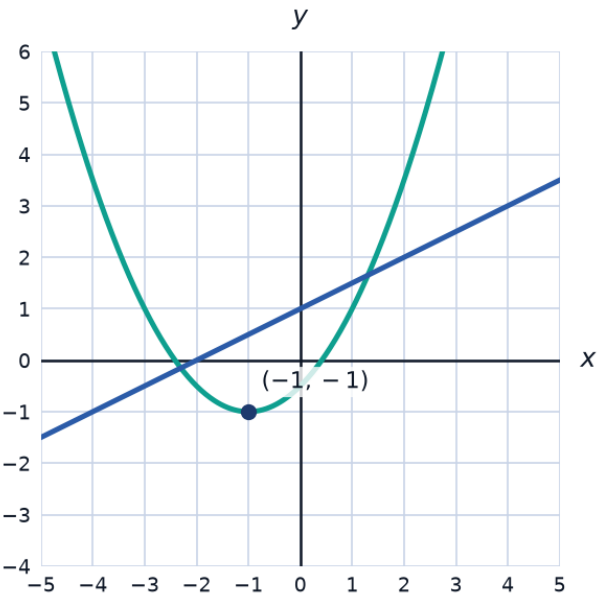
c. $a^2 = b^2 + c^2$

d. $c = a + b$

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28 [Foundational] • Target: Foci and Eccentricity of Hyperbolas

Find the foci of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.



a. $(\pm 5, 0)$

b. $(0, \pm 5)$

c. $(\pm\sqrt{7}, 0)$

d. $(\pm 25, 0)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Foci and Eccentricity of Hyperbolas

Find the eccentricity $e = c/a$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $5/4 = 1.25$

b. $4/5 = 0.8$

c. $3/4 = 0.75$

d. $\sqrt{7}/4$

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30 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

For $Ax^2 + Cy^2 + Dx + Ey + F = 0$ ($A, C \neq 0$), which condition identifies a hyperbola?

a. A and C have opposite signs ($AC < 0$)

b. A and C have the same sign and $A = C$

c. A and C have the same sign and $A \neq C$

d. $A = 0$ or $C = 0$

31 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

Identify the conic: $9x^2 - 4y^2 - 36x + 8y - 4 = 0$.

a. Hyperbola

b. Ellipse

c. Parabola

d. Circle

32 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

Convert $9x^2 - 4y^2 - 36x + 8y - 4 = 0$ to standard form.

a. $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$

b. $\frac{(x-2)^2}{9} - \frac{(y-1)^2}{4} = 1$

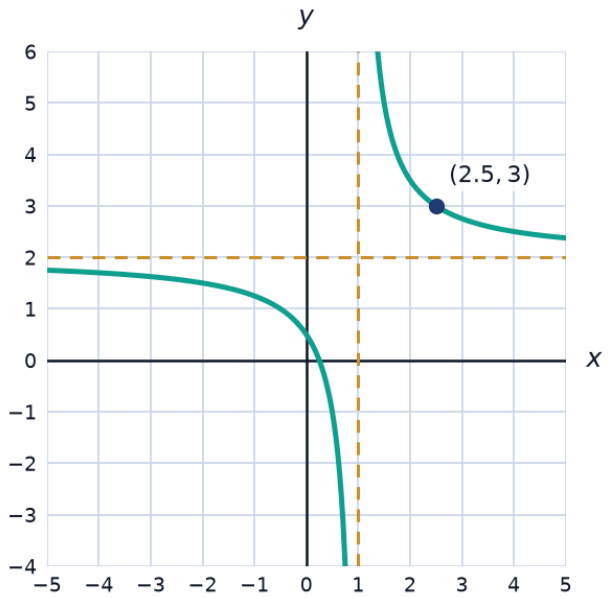
c. $\frac{(y-1)^2}{9} - \frac{(x-2)^2}{4} = 1$

d. $(x-2)^2 - (y-1)^2 = 36$

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33 **[Intermediate]** • **Target: Identifying Conics from General Second-Degree Equations**

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.



a. $y = \pm\frac{5}{3}x$

b. $y = \pm\frac{3}{5}x$

c. $y = \pm 15x$

d. $y = \pm\frac{8}{5}x$

34 **[Intermediate]** • **Target: Standard form**

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{5}{4}x$

b. $y = \pm 20x$

c. $y = \pm\frac{9}{5}x$

d. $y = \pm\frac{4}{5}x$

35 **[Intermediate]** • **Target: Identifying conics**

Find the asymptotes of the hyperbola $\frac{x^2}{36} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{1}{2}x$

b. $y = \pm 18x$

c. $y = \pm 2x$

d. $y = \pm\frac{3}{2}x$

36 **[Intermediate]** • **Target: Hyperbolas centered away from the origin**

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

a. $y = \pm 6x$

b. $y = \pm\frac{2}{3}x$

c. $y = \pm\frac{3}{2}x$

d. $y = \pm\frac{5}{3}x$

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37 [Intermediate] • Target: Eccentricity and applications

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{4}{3}x$

b. $y = \pm 12x$

c. $y = \pm\frac{7}{4}x$

d. $y = \pm\frac{3}{4}x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 1.03 Average Rate of Change

Find the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$.

a. $y = \pm\frac{1}{2}x$

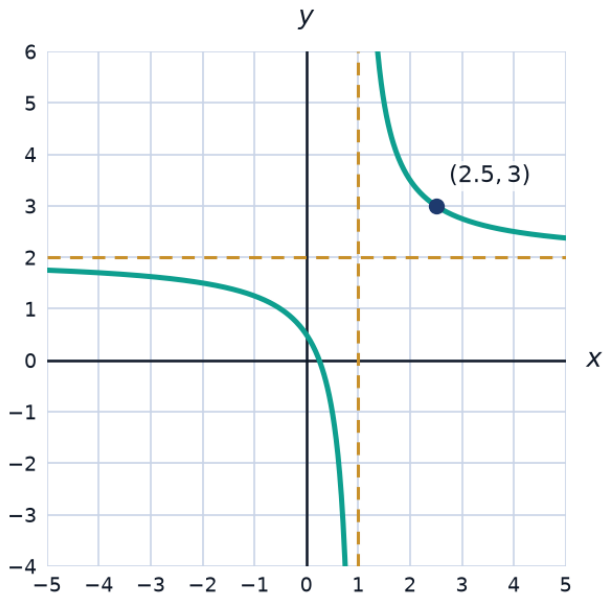
b. $y = \pm 2x$

c. $y = \pm\frac{3}{2}x$

d. $y = \pm -\frac{1}{2}x$

39 [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{1} = 1$.



a. $y = \pm 3x$

b. $y = \pm\frac{1}{3}x$

c. $y = \pm\frac{4}{3}x$

d. $y = \pm -\frac{2}{3}x$

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40 [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{1} = 1$.

a. $y = \pm -\frac{4}{5}x$

b. $y = \pm\frac{6}{5}x$

c. $y = \pm 5x$

d. $y = \pm\frac{1}{5}x$

41 [Challenging] • Target: Finding the equation from asymptotes and a point • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the asymptotes of the hyperbola $\frac{x^2}{36} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{2}{3}x$

b. $y = \pm 24x$

c. $y = \pm\frac{5}{3}x$

d. $y = \pm\frac{3}{2}x$

42 [Challenging] • Target: Applications: LORAN navigation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm 12x$

b. $y = \pm\frac{7}{3}x$

c. $y = \pm\frac{3}{4}x$

d. $y = \pm\frac{4}{3}x$

43 [Challenging] • Target: Standard Equations of Hyperbolas • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$.

a. $y = \pm\frac{1}{2}x$

b. $y = \pm 8x$

c. $y = \pm\frac{3}{2}x$

d. $y = \pm 2x$

44 [Challenging] • Target: Asymptotes of Hyperbolas • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{4} = 1$.

a. $y = \pm 10x$

b. $y = \pm\frac{5}{2}x$

c. $y = \pm\frac{7}{5}x$

d. $y = \pm\frac{2}{5}x$

45 [Challenging] • Target: Foci and Eccentricity of Hyperbolas • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{1} = 1$.

a. $y = \pm\frac{1}{4}x$

b. $y = \pm\frac{5}{4}x$

c. $y = \pm -\frac{3}{4}x$

d. $y = \pm 4x$

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